

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Tech.

Semester :VII

Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/

Electrical & Electronics Engg. / Electrical & Power Engineering

Subject Code & Name: BTEEOE704B Biomedical Instrumentation

Max Marks: 60

Date: 08/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Illustrate the function of the kidney with a neat schematic. Explain how blood flow takes place in the human body.	basic	6
B) Elucidate the basic mechanics of the spinal column and limbs with a necessary schematic.	understand	6
C) Explain the different components of a of a cell with the help of a neat schematic of it.	understand	6
Q.2 Solve Any Two of the following.		12
A) What is a blood gas analysers? Elucidate it with a neat schematic.	understand	6
B) Write a note on following terms: a) Heart rate b) Cardiac output	basic	6
C) Give a brief explanation of the erythrocyte sedimentation rate (ESR) using a neat diagram.	understand	6
Q. 3 Solve Any Two of the following.		12
A) Differentiate micro, needle and surface electrodes and state their applications.	basic	6
B) Explain the method for conducting an electromyography (EMG) using a block diagram.	understand	6
C) Derive an output voltage equation for a differential amplifier. State the significance of differential amplifier in biomedical instrumentation.	evaluate	6
Q.4 Solve Any Two of the following.		12
A) Demonstrate how ultrasonography (USG) can be utilized to diagnose abnormalities in the human body through a clear diagram.	understand	6
B) Elucidate briefly the different types of biotelemetry systems used currently in the medical field.	evaluate	6

- | | | | |
|---|---|------------|-----------|
| C) | Explain the procedure to analyze digital images. Illustrate the significance of biomedical instruments in analyzing digital images. | basic | 6 |
| Q. 5 Solve Any Two of the following. | | | 12 |
| A) | Explain the working of the ICCU patient monitoring system with a neat schematic. | understand | 6 |
| B) | Describe the advanced surgical techniques used by medical professionals worldwide. Explain in brief its significance for the current healthcare system. | understand | 6 |
| C) | Write a short note on the following terms: | basic | 6 |
| | a) Lithotripsy | | |
| | b) Dialysers. | | |

***** End *****